

PHAN VAN BANG

AI Engineer

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SUMMARY

AI Engineer with 2+ years of professional experience specializing in Multi-Agent Systems, RAG pipelines, Voice AI, and IoT integration. Built production-grade AI platforms including a multi-agent orchestration system with 94+ MCP tools, Swarm Intelligence, and continual learning capabilities.

- > **Short-term Goal:** Contribute as an AI Engineer at a product-driven company, applying expertise in Multi-Agent Systems and RAG pipelines to build scalable AI solutions that directly impact business metrics.
- > **Long-term Goal:** Evolve into a Senior AI Engineer / Tech Lead role, driving AI architecture decisions, mentoring team members, and contributing to open-source AI communities.

TECHNICAL SKILLS

Languages:	Primary: Python, TypeScript/JavaScript Proficient: C#, Java, Go Familiar: Rust, C++, C
AI & LLM:	LLM, Context Engineering (Hermes), Fine-tuning (LoRA, QLoRA), RAG & GraphRAG, Multi-Agent Orchestration, Swarm Intelligence, Continual Learning (XSkill), MCP Protocol (94+ tools), ComfyUI Workflows, HuggingFace, vLLM
Voice AI:	OpenAI Whisper STT, TTS (OpenAI/Fish Audio/ElevenLabs), VAD, Full-Duplex Voice Chat
Frameworks:	LangChain, LlamaIndex, FastAPI, Streamlit, React, React Native, Next.js, Express.js, Spring Boot, Laravel
Databases:	PostgreSQL + pgvector, Vector DB (Qdrant, Faiss, Chroma), SQL (MySQL, SQLite), NoSQL (MongoDB, Firebase), Graph DB (Neo4j)
Architecture:	Microservices, RESTful API, WebSocket, SSE Streaming, Docker Compose, Celery + Redis
DevOps & Cloud:	AWS (S3, EC2), Docker Hub, Cloudflare, GitHub Actions, Vercel, VPS Management, Nginx
AI IDEs & Tools:	Cursor, Claude Code, Codex, Roo Code (Custom Modes)
Tools:	Git, N8N Automation, Data Crawling/Scraping, Playwright, Unit Testing, PaddleOCR

WORK EXPERIENCE

AI Engineer MekongAI

06/2023 - Present

- > Designed & built multi-agent orchestration system (MekongAI Nexus) with 94+ MCP tools, Swarm Intelligence (debate, consensus, hierarchical, parallel), and XSkill continual learning, serving 500+ daily conversations
- > Developed AI learning content platform (Knowlympic) with OCR pipeline, automated question generation, reducing content creation time by 80% for 3 English language centers
- > Built AI interaction platform (MetaHuman) with personality engine, 3-tier memory system, full-duplex voice, and 3D avatar integration, achieving <500ms voice response latency
- > Integrated AI with ESP32 hardware for voice-interactive IoT bots via MCP protocol
- > Implemented GraphRAG memory with pgvector + hybrid retrieval (vector + FTS + graph RRF), improving retrieval accuracy by 35% over baseline vector search
- > Designed 13 security layers (CommandGuard, SSRF Guard, PII Redaction, etc.)
- > Built multi-channel deployment: WebSocket, Telegram, Discord, Slack, WhatsApp, Email, handling 1000+ concurrent connections
- > Developed RAG-based chatbots and agentic AI systems for enterprise clients
- > Fine-tuned LLM models using LoRA/QLoRA for domain-specific tasks
- > Managed VPS infrastructure, Docker deployments, and N8N automation workflows
- > Built 20+ ComfyUI AI image processing workflows, processing 5000+ images/month for clients
- > Collaborated in a team of 4-6 engineers, conducting code reviews and leading technical design discussions

KEY PROJECTS

1. MekongAI Nexus — Multi-Agent Orchestration System

- > Self-spawning agent system with Swarm Intelligence (4 strategies: debate, consensus, hierarchical, parallel)
- > 94 MCP tools across 28+ categories, GraphRAG memory, Deep Analysis Engine
- > XSkill continual learning (auto-extracts Skills + Experiences from trajectories)
- > 13 security layers, 9 communication channels, Docker Hub deployed

Tech: FastAPI, React, PostgreSQL + pgvector, Redis, Multi-Provider LLM, Docker

2. MekongAI Knowlympic — AI Learning Content Platform

- > AI-powered platform auto-generating learning content from textbook PDFs
- > Hybrid OCR (PaddleOCR + Gemini Vision), AI question generation/extraction
- > Auto test generator (multi-version A/B/C/D), RAG-based AI chat assistant

Tech: FastAPI, React, Celery + Redis, Qdrant, OpenAI, PaddleOCR

3. MekongAI MetaHuman — AI Personality Platform

- > AI with personality engine (5D Drive System, Emotion Critic, Inner Monologue)
- > 3-Tier Memory (Hot/Warm/Cold) with Ebbinghaus Forgetting Curve
- > Full-duplex voice conversation, 3D avatar with 10 emotion states

Tech: FastAPI, React, PostgreSQL, OpenAI Whisper/TTS, Fish Audio

4. ESP32 AI Bot — IoT Voice Assistant

- > Voice-interactive AI bot on ESP32 hardware with MCP protocol integration
- > Voiceprint recognition, visual perception, IoT device control via MQTT
- > Streaming ASR + LLM + TTS pipeline

Tech: Python, ESP32-S3, MQTT, MCP, FunASR, Multi-LLM

EDUCATION

B.Eng. in Information Technology

Da Nang, 2020 - 2025

Vietnam-Korea University of Information & Communication Technology (VKU)

CERTIFICATIONS & LANGUAGES

- > English: Professional working proficiency (Reading technical documentation, Writing, Speaking)
- > Self-study certifications: Deep Learning Specialization concepts, LangChain & LLM Application Development

SOFT SKILLS

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|---|---|---|
| > Self-learning & Rapid adaptation | > Clear technical communication & documentation | > Problem decomposition & analytical thinking |
| > Team collaboration & technical leadership | > Ownership mindset & deadline-driven execution | |